

ARTHUR W. TURNER

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PROFESSIONAL SUMMARY

AI Application Engineer with hands-on experience designing, building, and deploying GenAI systems, including RAG pipelines, LLM orchestration, multi-agent workflows, and evaluation frameworks. Builds production-oriented applications in Python using LangGraph, FastAPI, and cloud infrastructure, with a focus on retrieval quality, hallucination mitigation, and grounded responses. Every project in this portfolio represents individual contribution: concrete architecture decisions, working code, and measurable outcomes. AWS-certified across seven credentials including Generative AI Developer – Professional and Machine Learning – Specialty. Ph.D. with applied statistical modeling background. Local to the Westlake, TX area.

GENAI APPLICATION ENGINEERING

Intelligent Retrieval & Advisor-Assist System | Python · LangGraph · FastAPI · FAISS · ChromaDB

RAG-powered decision-support application • 2025

- Designed and implemented a multi-node RAG application in LangGraph with conditional routing across five query types (factual, procedural, synthesis, summarize, unsupported), embedding-based vector search, and configurable retrieval strategies optimized for answer relevance and retrieval precision.
- Engineered a hallucination mitigation system combining token-overlap grounding verification, retry/refuse loop, and dual confidence thresholds ensuring response faithfulness to source documents rather than LLM confabulation.
- Built a provider-agnostic LLM integration layer (OpenAI, Anthropic Claude, AWS Bedrock, Ollama) with FAISS/ChromaDB vector stores switchable via config enabling rapid A/B evaluation of retrieval quality and answer fidelity across providers.
- Implemented an evaluation framework with groundedness, relevance, and refusal-correctness scorers making retrieval quality and answer faithfulness auditable and reproducible.

Multi-Agent Orchestration Platform | Python · OpenAI · Tavily · Pydantic

Agent-based workflow with tool execution, routing, and task planning • 2024

- Architected a 4-agent pipeline (TopicSplitter → parallel Researchers → Synthesizer → Critic) using OpenAI function calling with concurrent ThreadPoolExecutor execution demonstrating agent-based task decomposition, parallel tool execution, routing, and result aggregation.
- Implemented a self-critique revision loop where the Critic agent evaluates synthesized outputs against factual consistency, citation accuracy, and logical coherence, then feeds structured revision instructions back, automated QA via agentic feedback, backed by Pydantic-validated inter-agent data contracts.

Decision-Simulation Engine | Python · TypeScript · Next.js · FastAPI · PostgreSQL · Docker

Full-stack GenAI application with event sourcing and analytics • 2025

- Architected a containerized full-stack application (Next.js, FastAPI, PostgreSQL, Docker Compose) with scenario versioning (draft/published states), OpenAPI-documented APIs, and an admin analytics layer computing completion rate, drop-off, and choice distribution.
- Designed an event-sourced session model with deterministic replay and a custom lexer → parser → AST evaluator for safe conditional logic that eliminates code injection risk while maintaining dynamic content generation.
- Backed by a 3,200+ line test suite and Playwright E2E coverage. Deployed via Docker Compose with reproducible multi-service orchestration.

Data Ingestion & Content Generation Pipeline | Python · NLP · TF-IDF · scikit-learn · OpenAI · Gemini

Structured data integration from heterogeneous document sources • 2024

- Engineered a 6-stage pipeline that ingests unstructured documents (PowerPoint, PDF, RTF, DOCX), integrates structured and unstructured data sources via multi-format parsers with style-aware heading detection, and uses TF-IDF cosine similarity to align content across document types.
- Built a provider-agnostic LLM orchestration layer (OpenAI → Gemini → Grok) with priority-based fallback so a failed API call degrades gracefully providing production resilience patterns for cost and latency optimization across GenAI pipelines.

Fault-Tolerant ETL & Content Generation System | Python · TypeScript · Next.js · Claude API · Vercel

Production-grade AI pipeline with caching, rate limiting, and multi-format export • 2025

- Architected a 4-stage ETL pipeline (ingest → clean → LLM generate → export) with YAML-driven config, SHA256-keyed disk caching, configurable RPM/TPM rate limiting, exponential backoff retry, and checkpoint/resume eliminating redundant API calls and enabling mid-run recovery without reprocessing.

- Delivered multi-format export adapters (structured data API, mobile JSON API, and a Next.js PWA deployed to Vercel) from a single pipeline run, with Pydantic v2 cross-field validators catching malformed LLM responses at parse time.

PROFESSIONAL EXPERIENCE

Instructor / Adjunct Professor | GPISD · Dallas College · Tarrant County College

2007 – Present

- Designed curriculum, assessments, and instructional systems across multiple institutions. Experience translating complex requirements into structured, user-facing deliverables.
- Built and deployed multiple GenAI applications to automate content generation, assessment creation, and data analysis workflows validated with real users and real data.
- Developed data-driven intervention systems identifying at-risk individuals through analytical methods and designing targeted support plans.

Research Strategist | International Mission Board of the SBC

1998 – 2007

- Led international research initiatives across four countries, managing cross-functional distributed teams and strategic communication programs.
- Built predictive models and statistical analyses; designed SQL-based data pipelines supporting global operations.

CERTIFICATIONS

AWS Certified (7 credentials): AI/ML: Generative AI Developer - Professional · Machine Learning - Specialty · Machine Learning Engineer – Associate · AI Practitioner **Cloud/Data:** Data Engineer - Associate · Solutions Architect - Associate · Cloud Practitioner

TECHNICAL SKILLS

GenAI / RAG: LangGraph · LangChain · RAG pipeline design · embeddings · vector search (FAISS, ChromaDB) · retrieval tuning · prompt engineering · hallucination mitigation · grounding verification · evaluation frameworks · multi-agent orchestration · agentic workflows · tool execution · routing · memory

LLM Platforms: OpenAI API · Anthropic Claude API · Google Gemini · AWS Bedrock · Ollama · xAI Grok

Languages & Frameworks: Python · TypeScript · SQL · FastAPI · Next.js (App Router) · React · Pydantic v2 · SQLAlchemy · Alembic

Data Integration: Multi-format document parsing (PDF, RTF, DOCX, PPTX) · ETL pipelines · structured/unstructured data ingestion · TF-IDF · NLP · Pandas · NumPy · NLTK · sentence-transformers

Infrastructure & DevOps: AWS (SageMaker · Lambda · Bedrock · Glue · Redshift) · Docker · Docker Compose · Vercel · Git · CI/CD · REST APIs · OpenAPI · structured logging · 12-factor config · checkpoint/resume · monitoring · testing (pytest, Playwright)

ML Frameworks: PyTorch · scikit-learn

EDUCATION

Ph.D., Political Science — The University of Alabama Applied predictive modeling (survival analysis) and quantitative methods.

M.A., Political Science — The University of Alabama

B.A., Political Science & German — Jacksonville State University

SELECTED PUBLICATIONS

"Post Authoritarian Elections: Testing Expectations About 'First' Elections," *Comparative Political Studies*. Statistical modeling of patterns in post-authoritarian states.